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59200 Kuala Lumpur W.P. Kuala Lumpur, Malaysia.

# Certificate of Analysis

COA No:

## PRODUCT DETAILS

Name

Batch

Date

Cell Source and Passage Human Umbilical Cord (Wharton's Jelly), Passage 3

Storage Temperature 4 – 8°C

Viable Cell Count 100.7 x 10<sup>6</sup>

| TEST                           | SPECIFICATION                                                                                           | TEST RESULTS                | REMARKS                         |
|--------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------|---------------------------------|
| Infectious Diseases Screening  | HBsAg; HCV Ab; HIV1/2 Ag/Ab; CMV Ab; RPR                                                                | Not Detected / Non-Reactive | <b>QUALITY CONTROL APPROVED</b> |
| Morphology of Cells            | Spindle-like cells                                                                                      | Small spindle shaped cells  | <b>QUALITY CONTROL APPROVED</b> |
| Cell Viability                 | ≥90% ± 0.1 of requested cells                                                                           | Within range                | <b>QUALITY CONTROL APPROVED</b> |
| Sterility                      | No cloudiness of the test media                                                                         | Negative                    | <b>QUALITY CONTROL APPROVED</b> |
| Bacterial Endotoxin Test (LAL) | Endotoxin sensitivity 0.125 EU/mL                                                                       | Negative                    | <b>QUALITY CONTROL APPROVED</b> |
| Mycoplasma                     | Mean of absorbance <0.05                                                                                | Negative                    | <b>QUALITY CONTROL APPROVED</b> |
| Immunophenotyping              | Positive marker – CD73, CD90, CD105 (≥ 95%)<br>Negative marker – CD11b, CD19, CD34, CD45, HLA-DR (≤ 2%) | Within range                | <b>QUALITY CONTROL APPROVED</b> |

The cell product delivered by Malaysian Genomics Resource Centre Berhad is manufactured and tested for research purpose only. Product is processed in a facility operating under current Good Manufacturing Practice (cGMP) guidelines in accordance with PIC/S standards.

Authorised Signature:

Quality Control Department  
Date:

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## Supporting Information

### MORPHOLOGY OF CELLS

Figure 1: Mesenchymal stem cells as viewed under an inverted phase contrast microscope (10x).

### STERILITY TEST <sup>6, 8</sup>

Aerobic microorganisms are incubated in Soybean Casein Digest Medium at 20°C - 25°C for 14 days.

Facultative microorganisms are incubated in Fluid Thioglycollate Medium at 30°C - 35°C for 14 days.

### BACTERIAL ENDOTOXIN TEST (LIMULUS AMOEBOCYTE LYSATE, LAL) TEST <sup>3,6</sup>

Negative findings for Gram-negative bacteria in cell culture based on FDA Reference of Standard Endotoxin Guideline.

### MYCOPLASMA TEST <sup>4,6</sup>

Negative findings for mycoplasma species *M. hyorhinis*, *M. arginini*, *M. fermentans*, *M. orale*, *M. Pirum*, *M. hominis*, *M. salivarium* and *A. laidlawii*.

### IMMUNOPHENOTYPING

| Marker     | Positive |       |        | Negative (CD11b, CD19, CD34, CD45, HLA-DR) |
|------------|----------|-------|--------|--------------------------------------------|
|            | CD73+    | CD90+ | CD105+ |                                            |
| Result (%) |          |       |        |                                            |

Table 1: Flow cytometric analysis was performed to verify the characteristic surface marker expression profile of mesenchymal stem cells.

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